

CT-F401RE EVALUATION BOARD

Final Design Report

Team: CoreTrace

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Institution: Muthoot Institute of Technology and Science (MITS), Kerala

Competition: Mixed Traces PCB Design

Tool: KiCad

1. PROBLEM STATEMENT

Design a compact and functional STM32F401RE evaluation and development board inspired by the STM32 NUCLEO-F401RE reference design.

The board must provide essential facilities for programming, debugging, and general-purpose embedded development within a 100×100mm 2-layer PCB, designed entirely in KiCad.

2. PROPOSED SOLUTION

The CT-F401RE is a compact evaluation board built around the STM32F401RET6 Cortex-M4 microcontroller. The design prioritizes:

- Clean power delivery via USB Micro-B

- Full debugging capability via SWD
- Complete communication interfaces
- Accessible GPIO expansion headers
- Professional silkscreen labeling
- SMD components for compact layout

3. CIRCUIT ARCHITECTURE

The board is divided into functional blocks:

POWER SECTION

- USB Micro-B provides 5V VBUS input
- AMS1117-3.3 LDO regulates 5V to 3.3V
- 22 μ F bulk capacitors on input and output
- 100nF ceramic decoupling caps on both sides
- Power LED indicates 3.3V rail is active

MCU SECTION

- STM32F401RET6 LQFP-64 at board center
- 100nF decoupling on all 4 VDD pins
- 100nF on VBAT pin
- 2.2 μ F on VCAP1 for internal regulator
- Ferrite bead (120R) on VDDA/VREF+ supply
- 1 μ F + 10nF + 10nF on VDDA filtering

CLOCK SECTION

- 8MHz HSE crystal oscillator

- 20pF load capacitors on PH0 and PH1
- Crystal placed close to MCU for short traces
- Internal PLL scales to 84MHz system clock

USB SECTION

- USB Micro-B connector on board edge
- 22 Ω series resistors on D+ and D-
- 1.5k Ω pullup on D+ for Full Speed detect
- PA11 = USB_DM, PA12 = USB_DP

SWD SECTION

- 6-pin header (VCC/SWDIO/SWCLK/NRST/GND)
- PA13 = SWDIO, PA14 = SWCLK
- Allows external ST-Link programmer

COMMUNICATION SECTION

- UART: PA2(TX), PA3(RX) — 4-pin header
- SPI: PA4(CS), PA5(SCK), PA6(MISO),
PA7(MOSI) — 6-pin header
- I2C: PB8(SCL), PB9(SDA) — 4-pin header
- I2C pullup resistors 4.7k Ω on SCL and SDA

GPIO SECTION

- CN1: 3.3V, PA0, PA1, PA8, PA9, PA10,
PA15, PB1, PB2, GND
- CN2: 3.3V, PB3, PB4, PB5, PB6, PB7,
PB10, PB12, PB13, GND

- 18 GPIO pins total exposed

USER INTERFACE

- PWR LED (Red) — 3.3V power indicator
- USR LED (Green) — user controlled via PB0
- TX LED (Blue) — UART transmit indicator
- RX LED (Yellow) — UART receive indicator
- SW2: Reset button on NRST
- SW3: User button on PC13
- SW1: SPDT slide switch for BOOT0

4. PCB DESIGN

Board specifications:

- Size: 100 × 100mm
- Layers: 2 copper layers (F.Cu + B.Cu)
- Software: KiCad 10.0.5

Component packages:

- Resistors: 0402 SMD
- Capacitors (small): 0402 SMD
- Capacitors (bulk): 0805 SMD
- LEDs: 0805 SMD
- Ferrite bead: 0805 SMD
- Crystal: SMD 3225

Track widths:

- Signal traces: 0.2mm
- MCU power pins: 0.3mm
- 3.3V power: 0.5mm
- VBUS 5V: 0.8mm

PCB features:

- GND copper pour on both layers
- Decoupling caps placed next to MCU VDD pins
- Crystal placed close to PH0/PH1 pins
- USB connector on board edge
- GPIO headers on top and bottom edges
- All connectors and pins clearly labelled

MAJOR COMPONENTS

Component	Value	Package	Cost
STM32F401RET6	MCU	LQFP-64	₹450
AMS1117-3.3	LDO	SOT-223	₹15
Crystal	8MHz	SMD 3225	₹25
USB Micro-B	Connector	SMD	₹20
Capacitors	Various	0402/0805	₹40
Resistors	Various	0402	₹13
LEDs	4 colors	0805	₹20
Switches	3 types	SMD	₹28
Pin Headers	2.54mm	THT	₹29
Ferrite Bead	120R	0805	₹8

PCB 2-layer 100x100mm ₹150

Total estimated cost: ₹811

6. DESIGN CONSIDERATIONS

POWER INTEGRITY

- Each VDD pin has dedicated 100nF cap
- Bulk 22μF caps for low frequency stability
- Ferrite bead isolates analog VDDA supply
- VCAP1 = 2.2μF as per STM32F401RE datasheet

SIGNAL INTEGRITY

- 22Ω USB series resistors reduce EMI
- Crystal traces kept short and equal length
- USB D+ and D- routed as matched pair
- SWD traces kept away from USB lines

USABILITY

- SPDT switch for easy BOOT0 selection
- FLASH position = normal operation
- BOOT position = bootloader mode
- 10kΩ pulldown ensures safe default state
- All header pins labelled on silkscreen
- Board name and team name on silkscreen

COST OPTIMIZATION

- SMD 0402 resistors — cheapest package
- AMS1117 over more expensive LDOs
- Standard 2.54mm headers
- No unnecessary components added

7. TESTING AND VALIDATION

ERC Results:

- Errors: 0 
- Warnings: 0 

DRC Results:

- Errors: 0 
- Warnings: 1 

(Footprint mismatch on SW1 — cosmetic only, caused by silkscreen edit to fix board edge clearance. No electrical impact.)

- Unconnected nets: 0 

Design validation:

- All minimum requirements verified
- All nets connected and routed
- All connectors labelled on silkscreen
- BOM reviewed and costed

- Schematic reviewed against datasheet
- PCB reviewed against design rules

8. CHALLENGES

- Started with wrong MCU (STM32F103C8Tx)
and had to redesign around STM32F401RET6
- VCAP1 pin unique to F401RE — not present
on F103, required careful attention
- Crystal footprint mismatch between 3-pin
symbol and 4-pin SMD package — resolved
by using Crystal_GND3_GND4 symbol
- Boot switch silkscreen overlapped board
edge — resolved by editing footprint
silkscreen inward
- First PCB attempt deleted and redesigned
for cleaner placement and routing

9. FINAL OUTCOME

The CT-F401RE evaluation board successfully
meets all competition requirements:

- ✓ STM32F401RET6 as main MCU
- ✓ USB Micro-B power and communication
- ✓ AMS1117-3.3 voltage regulation
- ✓ Complete decoupling strategy

- ✓ SWD debug interface
- ✓ UART, SPI, I2C headers
- ✓ 18 GPIO pins exposed
- ✓ 4 LED indicators
- ✓ Reset and user buttons
- ✓ BOOT0 SPDT switch
- ✓ 100×100mm 2-layer PCB
- ✓ ERC clean — 0 errors
- ✓ DRC clean — 0 errors
- ✓ Professional silkscreen
- ✓ Complete documentation

GitHub Repository:

<https://github.com/navaneeth-linh/CoreTrace-CT-F401RE>

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